

Phase Interferometer Direction-Finding with Multi-Frequency Sparse Arrays

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Abstract—To address the angle ambiguity problem of phase interferometer direction-finding (DF), a virtual multi-baseline phase interferometer based on multi-frequency sparse arrays is proposed. The proposed method maps multi-frequency components of a broadband signal to a reference frequency, thereby constructing virtual baselines. Subsequently, by adopting the long-short baseline method and the coprime baseline method, the unambiguous DF is achieved. Compared to conventional multi-baseline phase interferometer DF methods, the proposed method can achieve DF of the broadband signal with fewer sensors, therefore, have superior estimation performance due to the weaker sensor mutual coupling. Extensive simulation experiments validate the feasibility and superiority of the proposed method.

Index Terms—Phase interferometer, multi-frequency signal, direction-finding, angle ambiguity, virtual baseline.

I. INTRODUCTION

Direction-finding (DF) plays a crucial role in the field of array signal processing due to its extensive applications in radar, wireless communication, navigation and other domains [1]-[4]. The phase interferometer is one of the typical DF methods, which has the advantages of high DF accuracy, rapid response capability, and compact structure [5]-[15].

The phase interferometer determines the direction-of-arrival (DOA) by utilizing the phase difference of signals as received by spatially separated sensors [5]. The single-baseline configuration [6], a classic form of phase interferometer, offers the advantage of requiring fewer receiving channels. Nevertheless, it fails to achieve a satisfactory trade-off between DF ambiguity and measurement accuracy. In contrast, multi-baseline phase interferometers can improve the probability of ambiguity resolution while maintaining high DF precision [7].

Conventional multi-baseline phase interferometers include the long-short baseline interferometer [8]-[10] and the coprime baseline interferometer [11]-[14]. The former features a simple structure, low computational complexity, and high

DF accuracy. However, it requires the short baseline to be shorter than half the wavelength of the incident signal, and the small sensor spacing induces strong mutual coupling, making it unsuitable for broadband DF systems. For the latter, different baseline lengths are coprime, and DF ambiguity is resolved via the remainder theorem [15]. Although this method overcomes the half-wavelength constraint on the shortest baseline, it requires the baseline ratios to be coprime, which restricts the sensors layout. Additionally, all multi-baseline interferometer algorithms share a drawback in that a large number of sensors, thereby increasing the burden of DF system.

Through multi-frequency operation, multiple virtual sensors can be constructed using a small number of physical sensors. Reference [16] first proposed the use of multi-frequency to fill the holes in sparse arrays, significantly increasing the degrees of freedom of the virtual arrays. However, this study employs spatial spectrum method for DF, and the high computational complexity substantially increases the burden of the DF system. Most subsequent related research has focused on innovations in multi-frequency sparse arrays design while ignoring the computational complexity of DF algorithms [17]-[21].

This paper proposes a multi-frequency virtual multi-baseline phase interferometer to address the limitations of existing methods. Considering the reception of broadband signal by a sparse array, the proposed method selectively synthesizes signals from different frequency sub-bands after channelization processing, thereby generating an equivalent response of a virtual array operating at a unified frequency. This method eliminates the need for a restricted field of view [22] and enables unambiguous DF with fewer sensors while also reducing mutual coupling effects among the sensors.

II. PHASE INTERFEROMETER DF PRINCIPLES

In this section, the fundamental principles of phase interferometer, existing challenges, and typical solutions are briefly

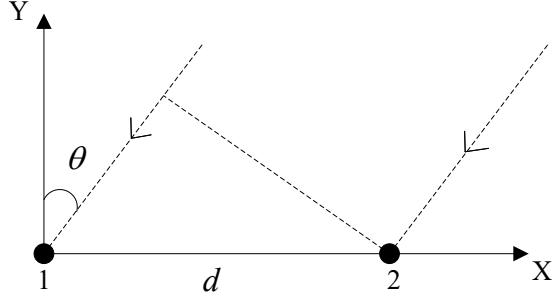


Fig. 1. Single-baseline array structure.

outlined.

A. Fundamental principles

Assuming a far-field source with a narrowband signal frequency of f_0 and a DOA of θ , the receiving array of a single-baseline phase interferometer consists of two array elements, as illustrated in Fig. 1. The true phase difference of the two received signals can be expressed as

$$\phi = 2\pi f_0 d \sin \theta / c, \quad (1)$$

where d denotes the baseline length and c is the propagation velocity of electromagnetic waves in the free space.

Based on Eq. (1), $\hat{\theta}$ can be directly derived as

$$\hat{\theta} = \arcsin \left(\frac{\phi c}{2\pi f_0 d} \right). \quad (2)$$

B. DF ambiguity

The phase difference between the received signals of the sensors is obtained through a phase detector, whose output range is confined to $[-\pi, \pi)$ [23]. When the true phase difference exceeds this range, the phase detector folds the true phase difference by 2π until the result lies within the output range, resulting in phase ambiguity.

From Eq. (1), when $d > c/(2f_0)$, for arbitrary θ , the true phase difference may exceed $[-\pi, \pi)$. The relationship between the true phase difference ϕ and the output result φ of the phase detector satisfies

$$\varphi = \phi - 2i\pi, i \in \mathbb{Z}. \quad (3)$$

The resulting angle is

$$\hat{\theta} = \arcsin \left(\frac{(\varphi + 2i\pi)c}{2\pi f_0 d} \right), i \in \mathbb{Z}. \quad (4)$$

C. DF ambiguity resolution

There are two methods to solve the DF ambiguity. The first approach is to limit the DF field of view to ensure that $i = 0$ in Eq. (4), meaning the true phase difference remains within the range $[-\pi, \pi)$ [22]. The second strategy is to employ multiple baselines, with the common methods including the long-short baseline method and the coprime baseline method.

However, all the aforementioned methods have certain limitations. For the limited field-of-view method, a relatively long

baseline is required to achieve high DF accuracy, which drastically restricts the unambiguous DF range and thus often fails to meet the requirements of practical engineering applications. In the long-short baseline method, the half-wavelength spacing becomes extremely small when the frequency of the signal is relatively high, rendering the physical installation of antennas impractical. Even if physical installation is feasible, the dense arrangement of sensors will induce strong mutual coupling effects. As for the coprime baseline method, it strictly maintains the coprime relationship among baseline lengths, which leads to the application limitation in the space-constrained scenarios.

III. MULTI-FREQUENCY VIRTUAL MULTI-BASELINE AMBIGUITY RESOLUTION METHOD

This section proposes a multi-frequency virtual multi-baseline method for phase interferometer to overcome the limitations of existing methods.

Assuming the signal received by the sparse array consists of M sensors is a broadband signal, after channelization, the frequency of the i -th signal is denoted as $f_i, i = 1, 2, \dots, L$, where L represents the total number of signals. The received signal can be expressed as [16]

$$\mathbf{X}(f_i) = \mathbf{A}(f_i, \theta) \mathbf{S}(f_i) + \mathbf{W}(f_i), \quad (5)$$

where $\mathbf{A}(f_i, \theta) \in \mathbb{C}^{M \times 1}$ denotes the array manifold matrix, $\mathbf{S}(f_i) = [s_1(f_i), s_2(f_i), \dots, s_N(f_i)] \in \mathbb{C}^{1 \times N}$ represents the N snapshots of the received signal, and $\mathbf{W}(f_i) = [\mathbf{w}_1(f_i), \mathbf{w}_2(f_i), \dots, \mathbf{w}_M(f_i)]^T \in \mathbb{C}^{M \times N}$ represents the noise vector from the M sensors.

Eq. (5) assumes that the sensors do not interfere with each other. In practice, any sensor output is influenced by its neighboring elements, which is called mutual coupling [24], [25]. Taking mutual coupling effects into account, Eq. (5) can be revised as follows

$$\mathbf{X}(f_i) = \mathbf{C} \mathbf{A}(f_i, \theta) \mathbf{S}(f_i) + \mathbf{W}(f_i), \quad (6)$$

where $\mathbf{C}$ represents the coupling matrix, which is only related to the sensor spacing, i.e., [24]

$$\langle \mathbf{C} \rangle_{n_1, n_2} = c_{|n_1 - n_2|}, |n_1 - n_2| \leq B, \quad (7)$$

where both n_1 and n_2 represent the element positions normalized with respect to the half-wavelength. Coupling coefficients $c_0, c_1, \dots, c_B$ satisfy $1 = c_0 > |c_1| > |c_2| > \dots > |c_B|$, and B represents the maximum element spacing.

Assuming the broadband signal satisfies the proportional spectrum assumption, the DOA remains constant across all selected frequency components and providing sufficient bandwidth to cover the required sub-bands [16]. To enable coherent multi-baseline synthesis, phase alignment is performed. Under the single-signal assumption, the relative phase offset between each sub-band and the reference frequency is estimated via the cross-covariance matrix of their channelization outputs. By extracting and averaging the phases of this matrix's diagonal elements, the initial phase differences are accurately determined and compensated.

To reduce the sensitivity of the phase interferometer to noise, the covariance matrix is calculated from the signal

matrix received by the physical array. By extracting the phases of the different elements according to the array configuration, the phase differences corresponding to various baselines can be determined [26].

For the single-baseline phase interferometer array structure shown in Fig. 1, the true phase difference is

$$\phi_i = 2\pi f_i d \sin \theta / c, i = 1, 2, \dots, L. \quad (8)$$

Let $f_i = \eta f_0, 0 < \eta \leq 1$, then

$$\phi_i = 2\pi \eta f_0 d \sin \theta / c, i = 1, 2, \dots, L, \quad (9)$$

where f_0 denotes the highest frequency point of the received signal, η represents the frequency point selection coefficient, and $d > c/(2f_0)$.

As indicated by Eq. (9), the selection of different frequency points enables the scaling of physical baselines, thereby constructing multiple virtual baselines. Thus, by appropriately setting η , a virtual multi-baseline structure (e.g., long-short baselines or coprime baselines) can be formed, which facilitates the achievement of high-accuracy and unambiguous DF.

A. Virtual long-short baseline method

Based on the principle of the long-short baseline method, the maximum baseline length for the short baseline is

$$d_{max} = \frac{c}{2f_0}. \quad (10)$$

Therefore, to construct a virtual short baseline, the selection range for η is

$$\eta \geq 1 - \frac{c}{2f_0 d}. \quad (11)$$

Let $\eta = 1 - c/(2f_0 d)$, then the phase difference of the short baseline is

$$\phi_s = \phi_l - 2\pi(1 - \frac{c}{2f_0 d})f_0 d \sin \theta / c = \pi \sin \theta, \quad (12)$$

where $\phi_l = 2\pi f_0 d \sin \theta / c$ is the theoretical unambiguous phase difference of the long baseline.

Therefore, the ambiguity integer for the phase difference of the long baseline can be expressed as

$$k = \text{round} \left(\left(\phi_s \frac{d}{d_{max}} - \hat{\phi}_l \right) / (2\pi) \right), \quad (13)$$

where $\text{round}(\cdot)$ denotes the rounding operator, and $\hat{\phi}_l$ is the output of the phase detector for the long baseline.

According to Eq. (13), the unambiguous phase difference of the long baseline can be reconstructed as follows

$$\bar{\phi}_l = 2\pi k + \hat{\phi}_l. \quad (14)$$

The obtained unambiguous DF result is

$$\hat{\theta} = \arcsin \left(\frac{\bar{\phi}_l c}{2\pi f_0 d} \right). \quad (15)$$

B. Virtual coprime baseline method

By referencing the principle of the virtual long-short baseline method, η is selected to construct virtual coprime baselines, where the physical baseline length d is an integer multiple of d_{max} .

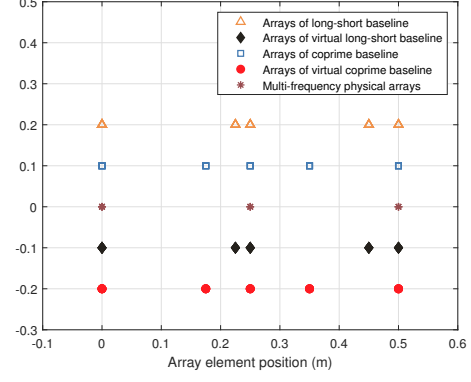


Fig. 2. Array geometric structures of diverse baselines.

The unambiguous phase differences for the virtual coprime baselines are respectively

$$\phi_1 = 2\pi f_0 \eta d \sin \theta / c, \quad (16)$$

$$\phi_2 = 2\pi f_0 d \sin \theta / c. \quad (17)$$

Assuming the output results of the two phase detector are $\hat{\phi}_1$ and $\hat{\phi}_2$ respectively, then

$$\phi_1 = \hat{\phi}_1 + 2\pi k_1, \quad (18)$$

$$\phi_2 = \hat{\phi}_2 + 2\pi k_2, \quad (19)$$

where k_1 and k_2 represent the ambiguity integer of $\hat{\phi}_1$ and $\hat{\phi}_2$, respectively.

Let $\eta = c_1/c_2$, where both c_1 and c_2 are integers and are coprime to each other. Substituting it into ϕ_1/ϕ_2 and rearranging yields

$$c_1 k_2 - c_2 k_1 = \frac{c_2 \hat{\phi}_1}{2\pi} - \frac{c_1 \hat{\phi}_2}{2\pi}. \quad (20)$$

Eq. (20) indicates that when k_1 and k_2 are solutions to this equation, $k_1 + nc_1$ and $k_2 + nc_2$ are also solutions, where n is an integer. Since c_1 and c_2 are coprime, a unique solution for k_1 and k_2 can be obtained.

IV. NUMERICAL RESULTS

This section presents the DF performance of the proposed methods, comparing them with the long-short baseline method and coprime baseline method (collectively referred to as conventional methods). The simulation experiments consist of two aspects. First, without considering the mutual coupling effects, comparing the DF accuracy and probability of ambiguity resolution under varying SNR. Second, it repeats the same comparison while accounting for mutual coupling effects. The comparison of DF accuracy is presented by the root mean square error (RMSE) of multiple unambiguous estimation results.

A. Comparison of DF performance under varying SNR in the absence of mutual coupling

Assuming the received broadband signal has a maximum frequency of 6GHz, with sufficient bandwidth, and the DOA is 60° . The position of sensors is set at $[0, 0.25, 0.5]$ m. The

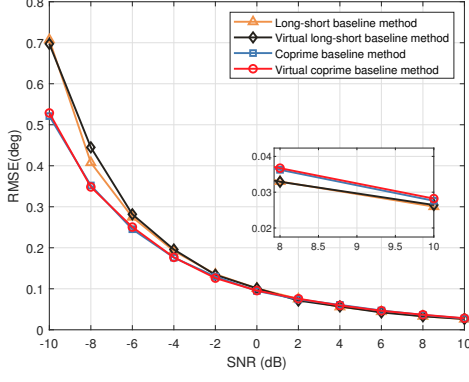


Fig. 3. RMSE versus SNR in the absence of mutual coupling.

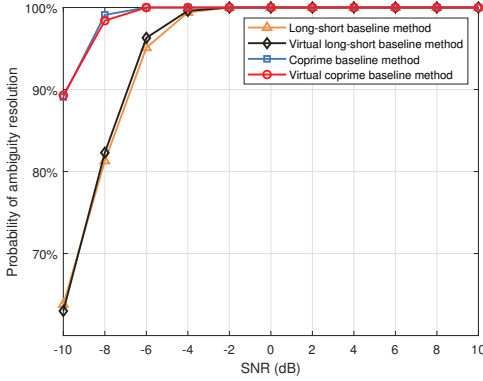


Fig. 4. Probability of ambiguity resolution versus SNR in the absence of mutual coupling.

SNR range is $(-10:2:10)$ dB, and the number of snapshots is $N=500$. For the virtual long-short baseline method, the baseline length for unambiguous DF is 0.025 m. Therefore, the frequency selection coefficient is setting as $\eta = 0.9$. For the virtual coprime baseline method, the baseline ratio is set to $7/10$, i.e., $\eta = 0.7$. For comparison, the long-short baseline method and coprime baseline method all employ five sensors to form the receiving array, with element positions are $[0, 0.225, 0.25, 0.45, 0.5]$ m and $[0, 0.175, 0.25, 0.35, 0.5]$ m, respectively. The specific array configurations are illustrated in Fig. 2. After 1000 Monte Carlo trials, the RMSE of the DF results and the probability of ambiguity resolution are shown in Fig. 3 and Fig. 4, respectively.

As shown in Fig. 3 and Fig. 4, across the entire range of SNR, the DF accuracy and ambiguity resolution probability of the proposed methods are essentially consistent with those of the corresponding conventional methods. These observations demonstrate that, in the absence of mutual coupling, the proposed methods can achieve DF performance comparable to that of conventional methods while using fewer sensors.

B. Comparison of DF performance under varying SNR in the presence of mutual coupling

In this experiment, the basic simulation conditions remain consistent with those in subsection A, with the exception that

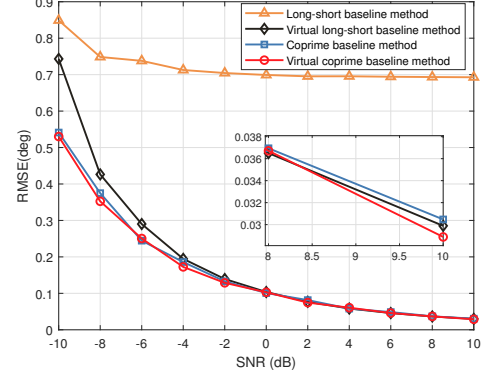


Fig. 5. RMSE versus SNR in the presence of mutual coupling.

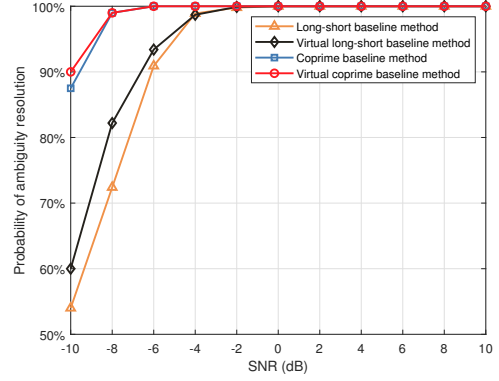


Fig. 6. Probability of ambiguity resolution versus SNR in the presence of mutual coupling.

a mutual coupling matrix is introduced. The elements of the mutual coupling matrix are defined as $c_1 = 0.3e^{j\pi/3}$, $c_l = c_1 e^{-j(l-1)\pi/3}/l$, $2 \leq l \leq B$ [24], where $B = 20$. Based on 1000 Monte Carlo trials, the RMSE of the DF results and the probability of ambiguity resolution are shown in Fig. 5 and Fig. 6, respectively.

Figs. 5-6 show that mutual coupling significantly degrades the conventional long-short method's accuracy and ambiguity resolution due to its dense sensor spacing. Conversely, the proposed virtual methods remain robust. While the conventional coprime method performs comparably to the proposed method due to its larger spacing, it requires more physical sensors, highlighting the practical superiority of the proposed method.

V. CONCLUSION

This paper proposes a multi-frequency virtual multi-baseline phase interferometer to achieve unambiguous DF with fewer sensors. By synthesizing virtual baselines, the method effectively mitigates mutual coupling and reduces hardware burden. Simulation results demonstrate that the proposed method achieves accuracy and ambiguity resolution comparable to conventional methods, while providing superior performance in the presence of mutual coupling. These advantages make it highly suitable for space-constrained and broadband DF applications.

REFERENCES

- [1] W. Zhou and Y. Zhou, "Research on Interferometer Direction Finding Technology Based on Digital Beam forming," in *2022 7th International Conference on Signal and Image Processing (ICSIP)*, Suzhou, China, 2022, pp. 54-58.
- [2] H. E. Çamlıbel, T. Doğan and S. Tokgöz, "In-Flight Calibration and Testing of Phase Interferometry Based Direction Finding Performance for Airborne Radar Warning Receivers," in *2023 IEEE AUTOTESTCON, National Harbor, MD, USA*, 2023, pp. 1-4.
- [3] P. Chen, Z. Chen, B. Zheng and X. Wang, "Efficient DOA Estimation Method for Reconfigurable Intelligent Surfaces Aided UAV Swarm," *IEEE Transactions on Signal Processing*, vol. 70, pp. 743-755, 2022.
- [4] K. Xu, M. Xing, R. Zhang, H. E, M. Sha and W. Nie, "High-Accuracy DOA Estimation Algorithm at Low SNR Through Exploiting a Supervised Index," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 58, no. 4, pp. 3658-3665, Aug. 2022.
- [5] W. -Y. Horng, "An Efficient DOA Algorithm for Phase Interferometers," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 3, pp. 1819-1828, Jun. 2020.
- [6] Y. Wu and L. Niu, "Studies on circular array interferometer direction finding," in *2015 IEEE 6th International Symposium on Microwave, Antenna, Propagation, and EMC Technologies (MAPE)*, Shanghai, China, 2015, pp. 49-52.
- [7] E. Jacobs and E. W. Ralston, "Ambiguity Resolution in Interferometry," *IEEE Transactions on Aerospace and Electronic Systems*, vol. AES-17, no. 6, pp. 766-780, Nov. 1981.
- [8] K. R. Sundaram, R. K. Mallik and U. M. S. Murthy, "Modulo conversion method for estimating the direction of arrival," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 36, no. 4, pp. 1391-1396, Oct. 2000.
- [9] C. He, J. Chen, X. Liang, J. Geng, W. Zhu and R. Jin, "High-Accuracy DOA Estimation Based on Time-Modulated Array With Long and Short Baselines," *IEEE Antennas and Wireless Propagation Letters*, vol. 17, no. 8, pp. 1391-1395, Aug. 2018.
- [10] L. Ding, J. He, T. Shu and T. Truong, "A generalized phase sample space approach for DoA estimation and array configuration optimization in phase interferometers," *Signal Processing*, 238 (2026): 110155.
- [11] R. L. Goodwin, "Ambiguity-resistant three-and four-channel interferometers Naval Research Lab," Washington, DC, USA, No. NRL-8005, 1976.
- [12] S. Samanta and M. Chakraborty, "Cooperative Base Line Interferometer for SWaP Optimized Direction Finding Receivers," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 60, no. 1, pp. 280-290, Feb. 2024.
- [13] J. -H. Lee and J. -M. Woo, "Interferometer Direction-Finding System With Improved DF Accuracy Using Two Different Array Configurations," *IEEE Antennas and Wireless Propagation Letters*, vol. 14, pp. 719-722, 2015.
- [14] H. Liang, X. Wang and A. Farina, "Performance Analysis and Optimal Placement for Unambiguous DOA Estimation using Interferometry," in *2020 IEEE Radar Conference (RadarConf20)*, Florence, Italy, 2020, pp. 1-6.
- [15] H. Zhang, G. Jin, Q. Xu and Q. Yan, "Phase unwrapping with Chinese Remainder Theorem for multi-baseline Interferometric Synthetic Aperture Radar," in *2013 10th International Conference on Fuzzy Systems and Knowledge Discovery (FSKD)*, Shenyang, 2013, pp. 93-97.
- [16] E. BouDaher, Y. Jia, F. Ahmad and M. G. Amin, "Multi-Frequency Co-Prime Arrays for High-Resolution Direction-of-Arrival Estimation," *IEEE Transactions on Signal Processing*, vol. 63, no. 14, pp. 3797-3808, Jul. 15, 2015.
- [17] L. Poli, N. Anselmi, M. Carlin and P. Rocca, "Compressive Sensing technique for multi-frequency sparse linear array design," in *2015 9th European Conference on Antennas and Propagation (EuCAP)*, Lisbon, Portugal, 2015, pp. 1-3.
- [18] S. Zhang, A. Ahmed, Y. D. Zhang and S. Sun, "DOA Estimation Exploiting Interpolated Multi-Frequency Sparse Array," in *2020 IEEE 11th Sensor Array and Multichannel Signal Processing Workshop (SAM)*, Hangzhou, China, 2020, pp. 1-5.
- [19] S. Zhang, A. Ahmed, Y. D. Zhang and S. Sun, "Enhanced DOA Estimation Exploiting Multi-Frequency Sparse Array," *IEEE Transactions on Signal Processing*, vol. 69, pp. 5935-5946, 2021.
- [20] M. W. T. S. Chowdhury and Y. D. Zhang, "Unambiguous DOA Estimation Using Multi-Frequency Rational Sparse Arrays," in *2023 57th Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, CA, USA, 2023, pp. 1324-1328.
- [21] Y. D. Zhang and M. G. Amin, "Multi-Frequency Rational Sparse Array for Direction-of-Arrival Estimation," in *2023 International Symposium on Signals, Circuits and Systems (ISSCS)*, Iasi, Romania, 2023, pp. 1-4.
- [22] Tuncer, T. E. , and B. Friedlander, "Classical and Modern Direction-of-Arrival Estimation," Academic Press (2009).
- [23] F. Mao, H. Ma, Y. Jiao, D. Li and Z. Gao, "Analysis of the research status of phase interferometer deblurring," in *2021 IEEE International Conference on Artificial Intelligence and Industrial Design (AIID)*, Guangzhou, China, 2021, pp. 667-672.
- [24] C. -L. Liu and P. P. Vaidyanathan, "Super Nested Arrays: Linear Sparse Arrays With Reduced Mutual Coupling—Part I: Fundamentals," *IEEE Transactions on Signal Processing*, vol. 64, no. 15, pp. 3997-4012, 1 Aug. 1, 2016.
- [25] C. -L. Liu and P. P. Vaidyanathan, "Super Nested Arrays: Linear Sparse Arrays With Reduced Mutual Coupling—Part II: High-Order Extensions," *IEEE Transactions on Signal Processing*, vol. 64, no. 16, pp. 4203-4217, 15 Aug. 15, 2016.
- [26] Y. Han, S. Wu, K. Wang, "An Improved Successive Ambiguity-Resolving Method for Interferometer," *Transactions of Beijing institute of Technology*, vol. 31, pp. 1090-1094 +1099, Sep. 2011.